AS · Unit 2 · WBI12 Cells, Development, Biodiversity and Conservation

Development of organisms

A complex organism starts as a single cell and grows by dividing and specialising. This topic looks at stem cells, the range of cells they can give rise to, and the sources of stem cells that medicine can use, along with the ethical questions those sources raise.

1. What a stem cell is

A stem cell is an undifferentiated cell. It has not yet taken on the particular structure and job of a mature cell, but it can give rise to other cell types. It has two key properties. It can divide and renew itself, so that the stem cell population is maintained, and it can develop into one or more specialised cell types. When a stem cell divides, one daughter may remain a stem cell while the other differentiates into a specialised cell. This is how the body keeps a supply of cells to replace worn-out or damaged tissue.

KEY IDEA

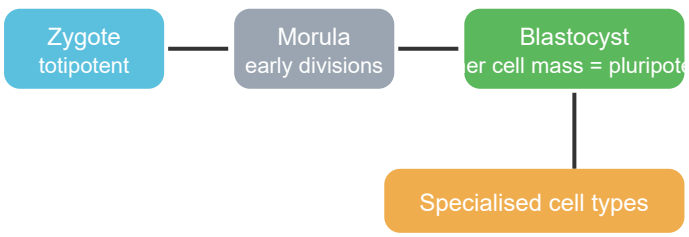
A stem cell is defined by two things: it is undifferentiated, and it can both renew itself and produce specialised cells.

2. Potency: totipotent and pluripotent cells

Potency describes how wide a range of cell types a stem cell can form. The more types it can form, the higher its potency.

A totipotent cell can form every cell type in an organism, including the cells of the placenta and umbilical cord, which lie outside the embryo itself. The zygote, made by the fusion of gametes, is totipotent, and so are the cells produced by the first few divisions of the fertilised egg.

A pluripotent cell can form almost all the cell types of the body, but not the extra-embryonic tissues such as the placenta. The inner cell mass of the blastocyst is pluripotent. Its cells are descended from the totipotent cells of the earlier divisions.



Property	Totipotent	Pluripotent
Where found	Zygote and first few divisions	Inner cell mass of the blastocyst
Cell types formed	All, including placenta and umbilical cord	Almost all embryonic and adult cell types, but not placenta
Relationship	Ancestor of pluripotent cells	Descended from totipotent cells

3. Early development: morula and blastocyst

After fertilisation the zygote divides by mitosis several times without growing in overall size. The result is a morula, a solid ball of cells. Continued division then produces a blastocyst. This is a hollow ball of cells with a fluid-filled cavity. The outer layer, the trophoblast, will form the placenta, and the inner cell mass sits at one end. It is this inner cell mass that is the source of pluripotent embryonic stem cells.

4. How stem cells give rise to specialised cells

Differentiation is the process by which a cell becomes specialised. The cell switches on some of its genes and switches off others, so it makes only the proteins that suit its job. A muscle cell makes large numbers of contractile proteins, and a nerve cell develops long extensions and the machinery to send electrical signals. Each cell still carries the full genetic information, so its specialisation comes from which genes are active rather than from losing genes.

As cells become more specialised they lose their ability to become other types. A pluripotent stem cell can form many cell types, but a cell that has committed to becoming a liver cell cannot turn into a nerve cell under normal conditions.

How a stem cell becomes a specialised cell

1. A stem cell contains the full set of genes, like every other cell in the body. Genes that every cell needs, such as those for respiration enzymes, are active, along with the genes that keep it unspecialised.
2. A signal, such as a chemical from neighbouring cells, causes some genes to be switched on and others to be switched off. Proteins called transcription factors bind to the DNA next to the genes that are switched on.

   This is differential gene expression: different cells use different parts of the same set of genes.
3. The genes that are switched on are transcribed, producing active mRNA, which leaves the nucleus through the nuclear pores.
4. The mRNA is translated on ribosomes, so the cell makes only the proteins coded for by its active genes.

5. These proteins control the cell's processes or build its structure. A muscle cell fills with contractile proteins, and a nerve cell grows long extensions and the machinery to send electrical signals.

6. The genes that are switched off are still present but are not transcribed, so their proteins are not made. As the cell becomes more specialised, it loses the ability to become other cell types.

This is why a liver cell cannot normally become a nerve cell, even though it still carries the genes for one.

Genes are switched on by transcription factors, proteins that bind to the DNA just in front of a gene so that it can be transcribed.

5. Sources of stem cells

Stem cells used in medicine can come from three main places. Each has its own advantages and drawbacks, and these shape which source is used for which treatment.

Embryonic stem cells

Embryonic stem cells are pluripotent and come from the inner cell mass of a blastocyst. They can be obtained from spare embryos left over from in-vitro fertilisation treatment, which are donated for research. A single cell can also be removed from an embryo at the eight-cell stage for genetic testing, and the rest of the embryo can still develop normally. Embryos can be grown until the blastocyst stage, and the inner cell mass is then cultured in the laboratory.

Their great advantage is their potency: they can form most human cell types and are easy to identify and isolate. Their main drawback is ethical. Using them destroys the embryo, and the supply of spare embryos is limited.

Adult stem cells

Adult stem cells are found in tissues of growing and mature bodies, for example in bone marrow, brain, liver and skin. They can only give rise to a limited range of cells within their own tissue. White blood cells, for example, can be formed from stem cells in bone marrow.

Adult stem cells raise fewer ethical concerns, because collecting them does not destroy an embryo. If they come from the patient themselves, the risk of rejection is very low. However, they are hard to identify and isolate, they divide poorly in culture, their potency is limited, and they can accumulate DNA changes as the person ages.

Umbilical cord blood stem cells

Blood taken from the umbilical cord and placenta after birth contains pluripotent stem cells. It is collected, the stem cells are multiplied in sterile culture, and the cord blood can be frozen and stored for later use. Its advantages are that it is more versatile than bone marrow cells, the chance of immune rejection is very small, and collection does

no harm to mother or baby. Its disadvantages are that the supply is limited, collection and storage are expensive, and a large storage facility is needed.

Source	Potency	Main advantage	Main disadvantage
Embryonic	Pluripotent	Forms most cell types and is easy to isolate	Ethically controversial; uses embryos; supply limited
Adult	Limited range within a tissue	Few ethical issues; low rejection risk for the same person	Hard to isolate; limited potency; short life in culture
Umbilical cord blood	Pluripotent	Very low rejection risk; no harm to mother or baby	Limited supply; costly to collect and store

COMMON MISTAKE

Do not assume adult stem cells are as flexible as embryonic ones. Adult stem cells are usually restricted to the tissue they come from, while embryonic and cord blood stem cells are pluripotent.

6. Using stem cells in medicine

Stem cells are used to replace cells that are lost or faulty. In Parkinson's disease, dopamine-producing nerve cells in the brain are gradually lost, and the lack of dopamine makes movement stiff and difficult. Experiments in monkeys have shown that embryonic stem cells can be coaxed into dopamine-producing neurons and transplanted, which improved symptoms. Diabetes, caused by destruction of the insulin-producing cells of the pancreas, is another target, because new cells could in principle restore insulin production. Damaged nerves in the brain and spinal cord may also be repaired using stem cells. In principle, stem cells from a patient could be grown into a tissue or an organ for transplant, avoiding a donor and reducing the risk of rejection.

KEY IDEA

Medical use depends on matching the source to the need. Pluripotency is useful when many cell types are required. A patient's own adult or cord blood stem cells are useful when rejection must be avoided.

7. Ethical decisions about stem cells

Society must weigh the benefits of stem cell research against concerns about the embryo. Regulating authorities decide how old an embryo can be when it is used for research, what counts as an acceptable source of stem cells, and how research is controlled to stop unnecessary repetition. They also ensure that cloning of humans is prevented. These judgements are made on ethical grounds, and they differ between countries.

Some people argue that an embryo has the potential to become a human being, so it should not be destroyed for research. Others argue that the potential benefit to patients justifies using spare embryos that would otherwise be discarded. Adult and umbilical cord sources avoid this objection, which is one reason they are often preferred when they are suitable.

8. Quick checks

QUICK CHECK

Which statement correctly describes a pluripotent stem cell?

- It can form every cell type, including the placenta
- It can only form cells within one tissue
- It can form almost all cell types of the body but not extra-embryonic tissue
- It has already committed to one specialised cell type

Pluripotent cells can form almost all body cell types but not the placenta or other extra-embryonic tissues, which is what separates them from totipotent cells. A cell limited to one tissue is an adult stem cell, and a committed cell is no longer a stem cell.

QUICK CHECK

Where are the pluripotent embryonic stem cells found in a blastocyst?

- In the outer trophoblast layer
- In the fluid-filled cavity
- Scattered through the placenta
- In the inner cell mass

The inner cell mass gives rise to the body of the embryo and is the source of pluripotent stem cells. The outer trophoblast forms the placenta, and the fluid cavity is not made of cells.

QUICK CHECK

Why are adult stem cells less ethically controversial than embryonic stem cells?

- They are always totipotent
- They can only be used on the same species
- Collecting them does not destroy an embryo
- They never need to be cultured

Adult stem cells are taken from tissues of a living person, so no embryo is destroyed. They are not totipotent, they can be used in other species in some research, and they usually do need culturing before use.

QUICK CHECK

Differentiation in a stem cell mainly involves which change?

- Losing its genes as it becomes specialised
- Switching to a different species of genetic code
- Switching some genes on and others off so only certain proteins are made
- Dividing faster than the stem cell next to it

A specialised cell keeps the full set of genes. Differentiation changes which genes are expressed, so the cell makes only the proteins its job needs. Division rate does not define differentiation.